INTRODUCTION

The aim of this practical session is to produce a data analysis report using chicwts R data set. This report explores the relationship between chicken weights and their feed.

Data

```
data(chickwts)

str(chickwts)
```

```
## 'data.frame':    71 obs. of  2 variables:
##  $ weight: num  179 160 136 227 217 168 108 124 143 140 ...
##  $ feed  : Factor w/ 6 levels "casein","horsebean",..: 2 2 2 2 2 2 2 2 2 2 ...
```

```
head(chickwts)
```

```
##   weight      feed
## 1    179 horsebean
## 2    160 horsebean
## 3    136 horsebean
## 4    227 horsebean
## 5    217 horsebean
## 6    168 horsebean
```

Results of the Data analysis

```
summary(chickwts)
```

```
##      weight             feed
##  Min.   :108.0   casein   :12
##  1st Qu.:204.5   horsebean:10
##  Median :258.0   linseed  :12
##  Mean   :261.3   meatmeal :11
##  3rd Qu.:323.5   soybean  :14
##  Max.   :423.0   sunflower:12
```

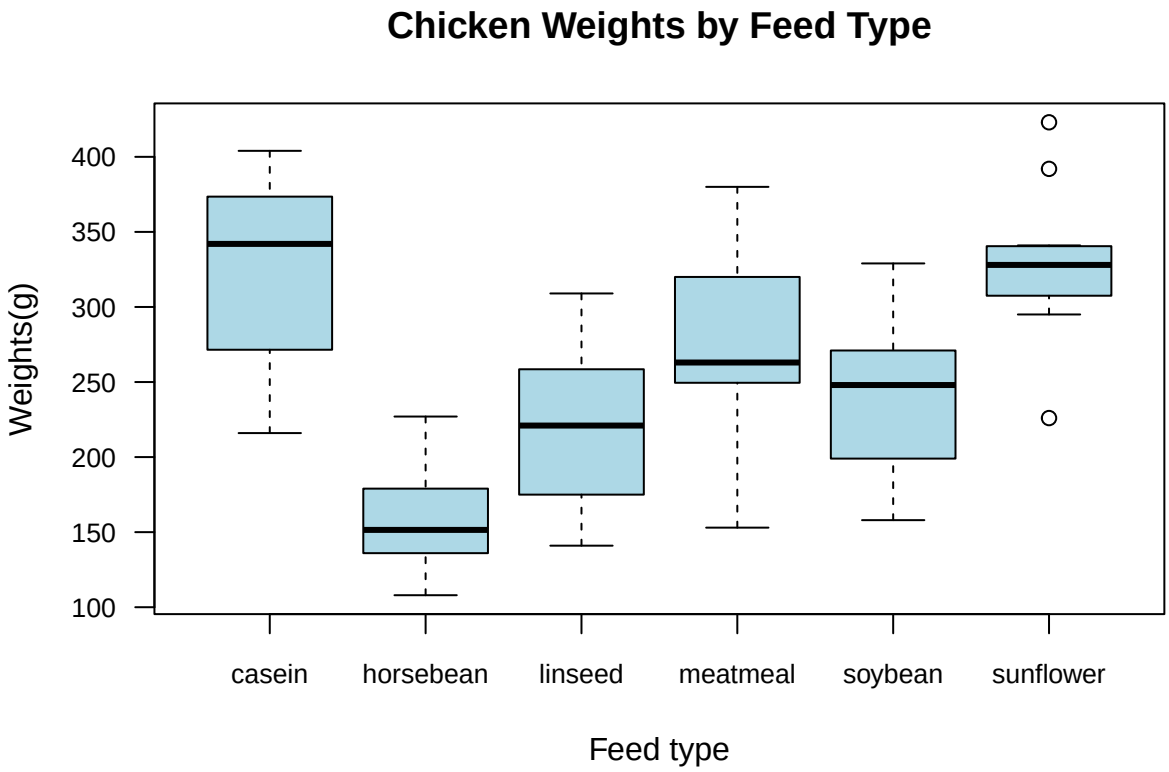
```
aggregate(weight~feed, data = chickwts, FUN=function(x){c(Mean=mean(x), SD=sd(x), Median=median(x))})
```

```
##        feed weight.Mean weight.SD weight.Median
## 1    casein   323.58333  64.43384     342.00000
## 2 horsebean   160.20000  38.62584     151.50000
## 3   linseed   218.75000  52.23570     221.00000
## 4  meatmeal   276.90909  64.90062     263.00000
## 5   soybean   246.42857  54.12907     248.00000
## 6 sunflower   328.91667  48.83638     328.00000
```

```r
#aggregate(weight ~ feed, data = chickwts, FUN = function(x) {
#  c(Mean = mean(x), SD = sd(x), Median = median(x))
#})
```

Graphical display

```r
boxplot(weight ~ feed, data = chickwts,
        main = "Chicken Weights by Feed Type",
        xlab="Feed type",
        ylab="Weights(g)",
        col="lightblue",
        las=1,
        cex.axis = 0.8)
```



Conclusion The descriptive statistics and the graphic representation clearly indicate that the type of feed affects chicken growth signficantly. Certain supplement feeds (like casein and sunflower) resulted in heigher average weights compared to others (such as horsebean and linseed).